Module 1: Fundamentals of Logical thinking – I

1. Number/Letter/Symbol Series

The goal is to identify the rule governing the sequence and use it to find the missing element.

A. Number Series

A Number Series is a sequence of numbers arranged in a specific, logical order based on a particular rule or pattern. Each number in the sequence is called a term.
The primary objective of problems based on number series is to identify the underlying rule that governs the formation of the sequence. Once the rule is determined, it can be used to:

- Find the next term(s) in the series.
- Find a missing term within the series.
- Sometimes, to identify the wrong term that does not follow the pattern**.**

Strategy: Follow a step-by-step approach.

1. Check for a **common difference** (subtraction).
2. If not, check for a **common ratio** (division).
3. Look for **squares, cubes, or other powers**.
4. Check if it's a **two-tier series** (the pattern is in the differences between terms).
5. Consider **recursive patterns** (each term depends on previous ones).
6. Look for **alternating patterns** (two different rules for odd and even positions).

Common Patterns with Examples:

1. Arithmetic (Linear) Series: Constant difference between terms.

- *Example 1:* 5, 8, 11, 14, 17, ?
  - +3 each time. **Next term: 20**
- *Example 2:* 25, 23, 21, 19, 17, ?
  - -2 each time. **Next term: 15**

2. Geometric (Exponential) Series: Constant ratio between terms.

- *Example 1:* 2, 6, 18, 54, ?
  - ×3 each time. **Next term: 162**
- *Example 2:* 128, 64, 32, 16, ?
  - ÷2 (or ×0.5) each time. **Next term: 8**

3. Perfect Square/Cube Series:

- *Square Series:* 1, 4, 9, 16, 25, ?
  - $1^2, 2^2, 3^2, 4^2, 5^2$,... **Next term: 36 (6^2)**
- *Cube Series:* 1, 8, 27, 64, ?
  - $1^3, 2^3, 3^3, 4^3$,... **Next term: 125 (5^3)**

4. Two-Tier Series: The difference between terms has its own pattern.

- *Example 1:* 2, 4, 7, 11, 16, ?
  - Differences: +2, +3, +4, +5
  - Next difference: +6

    - **Next term: 16 + 6 = 22**
- *Example 2:* 1, 2, 5, 10, 17, 26, ?
    - Differences: +1, +3, +5, +7, +9 (odd numbers)
    - Next difference: +11
    - **Next term: 26 + 11 = 37**

5. Recursive Series: A term is based on one or more previous terms.

- *Fibonacci Type:* Each term is the sum of the two previous terms.
    - *Example:* 3, 4, 7, 11, 18, 29, ?
    - 3+4=7, 4+7=11, 7+11=18, 11+18=29
    - **Next term: 18 + 29 = 47**
- *Other Recursive:* Each term is (previous term × n) + m.
    - *Example:* 3, 5, 11, 35, ?
        - (3 × 1) + 2 = 5
        - (5 × 2) + 1 = 11
        - (11 × 3) + 2 = 35
        - Pattern: ×1, +2 / ×2, +1 / ×3, +2...
        - **Next term: (35 × 4) + 1 = 141**

6. Alternating Series: Two series interwoven.

- *Example:* 2, 4, 3, 6, 5, 10, 9, ?
    - Split into odd and even positions:
        - Odd Positions (1st, 3rd, 5th, 7th): 2, 3, 5, 9 (

1. Number Series

Here the sequence is formed with **rules on numbers** such as:

- Addition/Subtraction
- Multiplication/Division
- Squares/Cubes
- Prime numbers
- Mixed rules

Examples:

Example 1:

5, 10, 15, 20, ?

Rule: +5 each time

Answer = **25**

Example 2:

2, 4, 8, 16, ?

Rule: Multiply by 2 each time

Answer = **32**

Example 3:

1, 4, 9, 16, 25, ?

Rule: Square numbers (1^2, 2^2, 3^2, 4^2, 5^2, …)

Answer = **36 (6^2)**

Example 4:

3, 6, 11, 18, 27, ?

Rule: Differences are +3, +5, +7, +9, … (odd numbers)

Answer = **38**

1. Simple Addition / Subtraction

Q1: 7, 14, 21, 28, ?

Rule: +7 each time

Answer = **35**

Q2: 100, 95, 90, 85, ?

Rule: -5 each time

Answer = **80**

2. Multiplication / Division

Q3: 2, 4, 8, 16, 32, ?

Rule: ×2 each time

Answer = **64**

Q4: 729, 243, 81, 27, 9, ?

Rule: ÷3 each time

Answer = **3**

3. Square / Cube Numbers

Q5: 1, 4, 9, 16, 25, ?

Rule: Squares of 1,2,3,4,5,…

Answer = **36 (6^2)**

Q6: 1, 8, 27, 64, 125, ?

Rule: Cubes of 1,2,3,4,5,…

Answer = **216 (6^3)**

4. Prime Numbers

Q7: 2, 3, 5, 7, 11, 13, ?

Rule: Prime numbers

Answer = **17**

5. Difference Pattern

Q8: 3, 6, 11, 18, 27, ?

Differences = 3, 5, 7, 9,… (odd numbers)

Answer = **38**

Q9: 5, 10, 19, 32, 49, ?

Differences = 5, 9, 13, 17 (increasing by +4)

Next difference = 21

Answer = **70**

6. Mixed Operations

Q10: 2, 4, 12, 48, ?

Rule: ×2, ×3, ×4, ×5 …

Answer = **240**

Q11: 1, 2, 6, 24, 120, ?

Rule: Factorials (1!, 2!, 3!, 4!, 5!, …)

Answer = **720 (6!)**

Letter Series

Letter series are based on the **position of alphabets** (A=1, B=2, … Z=26).

Patterns may involve **addition, subtraction, skipping letters, reverse order, or combinations**.

1. Simple Forward / Backward

Q1: A, B, C, D, ?

Rule: +1 each time

Answer = **E**

Q2: Z, Y, X, W, ?

Rule: -1 each time (reverse order)

Answer = **V**

2. Skipping Letters

Q3: A, C, E, G, ?

Positions: 1, 3, 5, 7… (+2 each time)

Answer = **I**

Q4: B, E, H, K, ?

Positions: 2, 5, 8, 11… (+3 each time)

Answer = **N**

3. Alternate Forward/Backward

Q5: A, Z, B, Y, C, ?

Rule: Forward +1, Backward -1 alternately

Answer = **X**

Q6: M, N, L, O, K, P, ?

Rule: One step forward, one step backward

Answer = **J**

4. Double / Repeated Letters

Q7: A, BB, CCC, DDDD, ?

Rule: Letter repeats as per its position

Answer = **EEEEE**

Q8: AA, CC, EE, GG, ?

Even letters repeated twice

Answer = **II**

5. Complex Patterns

Q9: A, C, F, J, O, ?

Positions: 1, 3, 6, 10, 15 (triangular numbers)

Answer = **U (21st letter)**

Q10: D, H, L, P, ?

Positions: 4, 8, 12, 16 (+4 each time)

Answer = **T**

- **Complex Letter Series Examples**

1. Increasing Gaps

Q1: A, C, F, J, O, ?

Positions: 1, 3, 6, 10, 15 (+2, +3, +4, +5 …)

Answer = **U (21st letter)**

2. Two Different Series Mixed

Q2: A, Z, B, Y, C, X, ?

Two series running:

- A, B, C … (forward)
- Z, Y, X … (backward)
- Answer = **D, W**

3. Odd–Even Position Rule

Q3: A, D, I, P, ?

Positions: 1, 4, 9, 16 … (Squares)

Answer = **Q (17th letter)**

4. Alternating Step Size

Q4: B, E, I, N, T, ?

Positions: +3, +4, +5, +6 …

Answer = **A (27 → wraps to 1st letter)**

5. Reversing Inside Series

Q5: Z, X, U, Q, L, ?

Positions: -2, -3, -4, -5 …

Answer = **F**

6. Repeated Pairs

Q6: A, A, B, B, C, C, D, D, ?

Rule: Each letter repeated twice

Answer = **E, E**

7. Vowel–Consonant Pattern

Q7: A, B, E, F, I, J, O, ?

Pattern: Vowel, Consonant, Vowel, Consonant …

Answer = **P**

8. Jumbled / Jumping Pattern

Q8: A, F, K, P, U, ?

Positions: 1, 6, 11, 16, 21 … (+5 each time)

Answer = **Z (26th letter)**

9. Combination of Forward + Reverse

Q9: M, N, L, O, K, P, J, ?

Rule: One step forward, one step backward alternately

Answer = **Q**

10. Using Multiplication of Positions

Q10: A, C, G, M, ?

Positions: 1, 3, 7, 13 (+2, +4, +6 …)

Answer = **S (19th letter)**

Odd Man Out

Definition (in simple terms)

Odd Man Out means: In a given sequence of numbers, letters, or symbols, **all follow a rule/pattern except one**. The task is to identify that one **different (odd) element**.

Odd man out means **all items follow the same rule except one**, and you have to **find the different one**.

1. Which number is the odd one out?

2, 4, 6, 7, 8, 10

Answer: 7

- **Why?** 2, 4, 6, 8, and 10 are all **even numbers**. 7 is an **odd number**.

2. Which number is the odd one out?

3, 6, 9, 12, 14, 15

Answer: 14

- **Why?** 3, 6, 9, 12, and 15 are all **multiples of 3**. 14 is not (14 ÷ 3 = 4.666...).

3. Which number is the odd one out?

5, 10, 15, 20, 23, 25

Answer: 23

- **Why?** 5, 10, 15, 20, and 25 are all **multiples of 5**. 23 is not.

4. Which number is the odd one out?

11, 13, 17, 19, 21, 23

Answer: 21

- **Why?** 11, 13, 17, 19, and 23 are all **prime numbers** (only divisible by 1 and themselves). 21 is a **composite number** (3 x 7 = 21).

5. Which number is the odd one out?

16, 25, 36, 48, 49

Answer: 48

- **Why?** 16 (4x4), 25 (5x5), 36 (6x6), and 49 (7x7) are all **perfect squares**. 48 is not a perfect square.

6. Which number is the odd one out?

1, 4, 9, 16, 25, 30

Answer: 30

- **Why?** 1 (1x1), 4 (2x2), 9 (3x3), 16 (4x4), and 25 (5x5) are all **perfect squares**. 30 is not a perfect square.

7. Which number is the odd one out?

10, 20, 30, 35, 40, 50

Answer: 35

- **Why?** 10, 20, 30, 40, and 50 are all **multiples of 10**. 35 is a multiple of 5, but not 10.

1. Find the odd one out.

A) Apple

B) Banana

C) Orange

D) Carrot

Answer: D) Carrot

- **Justification:** Apple, Banana, and Orange are fruits. Carrot is a vegetable.

2. Find the odd one out.

A) 2

B) 4

C) 6

D) 7

Answer: D) 7

- **Justification:** 2, 4, and 6 are even numbers. 7 is an odd number.

3. Find the odd one out.

A) Cat

B) Dog

C) Cow

D) Lion

Answer: D) Lion

- **Justification:** Cat, Dog, and Cow are common domestic animals. A Lion is a wild animal.

4. Find the odd one out.

A) Pen

B) Pencil

C) Eraser

D) Water

Answer: D) Water

- **Justification:** Pen, Pencil, and Eraser are things you use for writing. Water is not.

5. Find the odd one out.

A) Monday

B) Tuesday

C) Friday

D) January

Answer: D) January

- **Justification:** Monday, Tuesday, and Friday are days of the week. January is a month.

6. Find the odd one out.

A) Red

B) Blue

C) Green

D) Rose

Answer: D) Rose

- **Justification:** Red, Blue, and Green are colors. Rose is a flower (or a shade of a color, but primarily it's a thing, not a basic color name).

7. Find the odd one out.

A) △ Triangle

B) □ Square

C) ◯ Circle

D) ⬢ Pentagon

Answer: C) ◯ Circle

- **Justification:** A Triangle, Square, and Pentagon are shapes with straight sides. A Circle is round with no sides.

8. Find the odd one out.

A) Eye

B) Ear

C) Nose

D) Chair

Answer: D) Chair

- **Justification:** Eye, Ear, and Nose are parts of the face/body. A Chair is a piece of furniture.

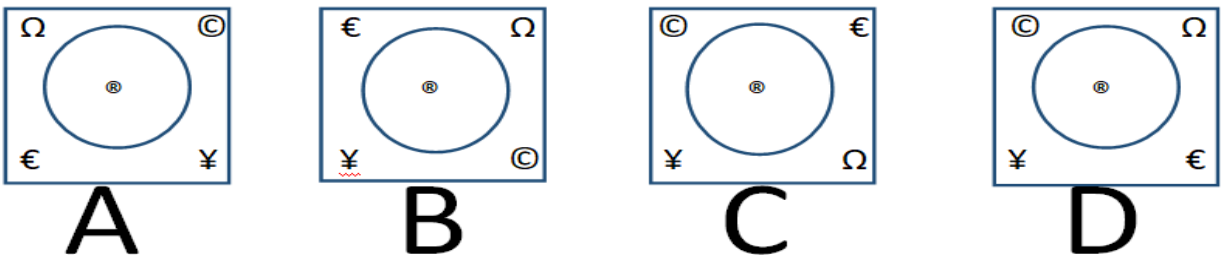
9. Which letter is the odd one out?

a, e, i, o, u, k

Answer: k

- Why? The letters a, e, i, o, u are called vowels. The letter k is a consonant.

10. Find the odd one out.



a) Figure A

b) Figure B

c) Figure C

d) Figure D

- Answer: c

Blood RELATION

1) If A is the brother of B, B is the sister of C, and C is the father of D, how is A related to D?

A) Uncle

B) Brother

C) Nephew

D) Cannot be determined

Ans: A) Uncle

2) Pick the right choice from the given options

If Esha is Mother of Karan, Karan is Grandson of Seema, Seema is Mother of Bharat, Bharat is Son of Sanjay, Sanjay is Father of Abhishek.

Esha is _____ of Abhishek

A) Granddaughter
B) Daughter
C) Wife
D) Mother
Ans: C

3) A is son of C while C and Q are the sisters to one another. Z is the mother of Q. If P is the son of Z, Which one of the following statements is correct?

A. Q is the grandfather of A ❌
B. **P is the maternal uncle of A** ✅
C. P is the cousin of A ❌
D. Z is the brother of C ❌

✓ Correct Answer: **B. P is the maternal uncle of A**

4) **A man said to a lady, "The son of your only brother is the brother of my wife." What is the lady to the man?**

Mother
B. Sister
C. Sister of father-in-law
D. Grandfathe

Answer: Option C

5. Introducing Rajesh, Neha said, “His brother’s father is the only son of my grandfather”. How Neha is related to Rajesh?

(a) Sister (b) Daughter (c) Mother (d) Niece

Correct Answer: A

6. Deepak said to Nitin, “That boy playing with the football is the younger of the two brothers of the daughter of my father’s wife”. How is the boy playing football related to Deepak?

(a) Son (b) Brother (c) Causin (d) Nephew

Correct Answer: b

7. A and B are married couple. X and Y are brothers. X is the brother of A. How is Y related to B?

   (a) Brother-in-law (b) Brother (c) Son-in-law (d) Cousin

Correct Answer: a

8. A is the son of C; C and Q are sisters; Z is the mother of Q and P is the son of Z. Which of the following statements is true?

(a) P and A are cousins
(b) P is the maternal uncle of A
(c) Q is the maternal grandfather of A
(d) C and P are sisters

Correct Answer: b

9. A man said to lady, “Your mother’s husband’s sister is my aunt.” How is the lady related to the man?

   (a) Daughter (b) Grand daughter (c) Mother (d) Sister

Correct Answer: d

10. X introduces Y saying, "He is the husband of the granddaughter of the father of my father ". How is Y related to X?

a	Son-in-law	
b	**Brother-in-law**	correct
b	Son	
d.	Brother	

answer b

Marks Awarded													
Marks After Verify-cation													Signature of the Verification Examiner

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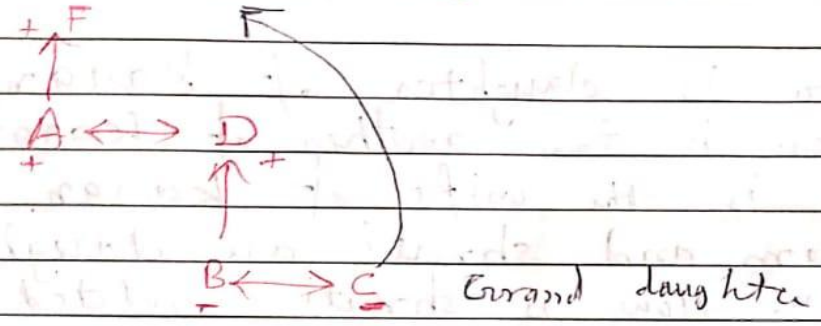
(Begin answer for each question on a new page)

(1) A is brother of D, D is father of B, B and C are sisters, F is the father of A then how C related to F



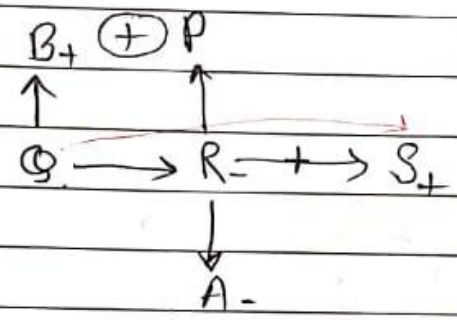
1) upper generation ↑ Mother, father
Grand father, Grand mother
uncle, Aunt

2) Our Generation, self, Brother, sister, cousins

3 Level same ⟷

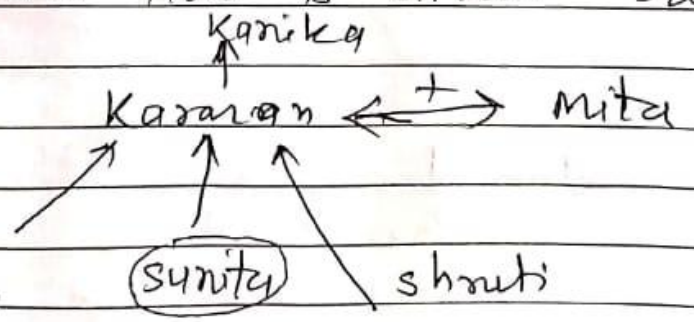
4) lower Generation son, Daughter, Nipher
Niece

(2) Q B is the father of Q. B has only two children. Q is the brother of R. R is the daughter of P. A is the grand daughter of P. S is the father of A. How is S related to Q.



1) son
2) son-in-law
3) Brother in-law
4) Brother in Law.

3) Q. Sunita is daughter of Karan, Kanika is the mother of Karan. Mita is the wife of Karan. Poonam and shruti are daughters of Karan. How is shruti related to Kanika



a) Mother
b) Grand-father
c) Ground ~~son~~ daughter
d) Ground mother.